[TODO: Title, e.g. Will It Fly? Closed-Loop Evaluation of Learned World Models for Cooperative Slung-Load Transport]

[TODO: First author]
[TODO: Department]
[TODO: Organization]
[TODO: City, Country]
[TODO: Email or ORCID]

[TODO: Second author]
[TODO: Department]
[TODO: Organization]
[TODO: City, Country]
[TODO: Email or ORCID]

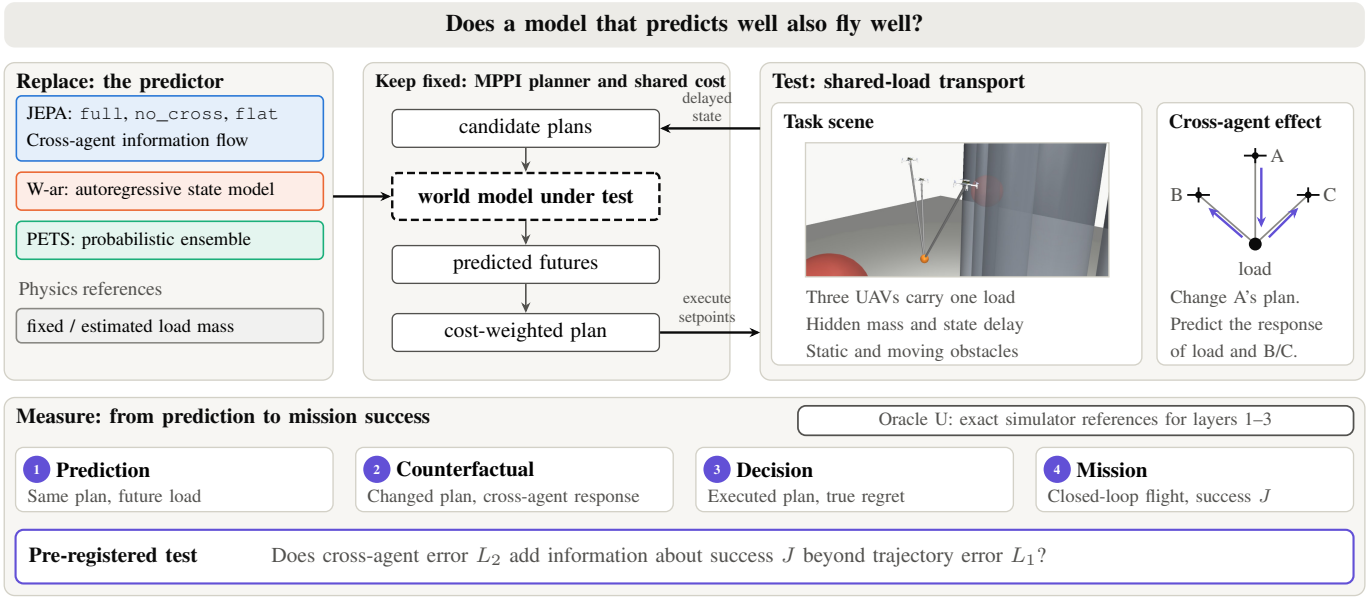


Fig. 1. World-model evaluation under a fixed planner and cost. Three quadrotors transport a slung load; only the predictor changes. Layers 1–3 use exact simulator references (Fig. 4); layer 4 measures closed-loop mission success. The pre-registered test asks whether cross-agent error adds information beyond trajectory error.

***Abstract*—World models are often selected by held-out prediction error, but planners use them to compare plans on the states they visit. For aircraft carrying a shared load, this requires predicting how one aircraft's move affects the load and its teammates. We introduce an open-source arena in which three Crazyflie 2 quadrotors transport a load of unknown mass past static and moving obstacles under delayed observations. A shared planner evaluates world models on four layers: open-loop prediction, cross-agent counterfactual prediction, decision quality and mission success, with exact simulator references for the first three. A pre-registered analysis across 30 trained model instances tests whether cross-agent error adds information about success beyond open-loop error. Open-loop error on the planner's states is the strongest single correlate of success; cross-agent error adds information, mainly through the contrast between PETS and JEPA `flat`. Decision quality correlates with success least. Physics with online mass estimation achieves a success rate similar to the oracle's, while the best learned models achieve rates similar to nominal-mass physics. These findings support evaluating team world models through complementary prediction diagnostics and closed-loop missions.**

***Index Terms*—[TODO: e.g. world models, model-based planning, multi-UAV systems, cable-suspended payload, closed-loop evaluation, benchmark.]**

I. Introduction

Learned dynamics models capture aerodynamic disturbances [1] and downwash between multirotors [2], [3], support long-horizon quadrotor control [4], and guide cooperative payload transport in simulation [5]. Inside a planner, a world model predicts the futures of candidate plans so that the planner can compare their costs.

Aviation controllers undergo closed-loop testing in simulation and with hardware before flight tests [6], [7]. Learned world models are commonly selected by held-out prediction

error, yet the planner may visit different states. In our development runs, a multilayer perceptron (MLP) had ten times its held-out error on the planner's states and completed fewer than one mission in five. Does a model that predicts well also fly well?

Prediction accuracy and control performance can diverge: one-step likelihood need not track reward [8], low prediction error can coexist with poor action choices [9], [10], and quadrotor predictors with errors an order of magnitude apart can achieve similar success under one planner [11]. Cooperative slung-load transport adds physical coupling [12]–[15]: one aircraft's move affects the load and the other aircraft through their cables. We ask whether predicting these cross-agent effects adds information about mission success beyond trajectory prediction error.

Our arena tests world models in a shared planner (Fig. 1). Three Crazyflie 2 quadrotors carry a load of unknown mass past static and moving obstacles, with observations delayed by up to 200 ms (Fig. 2). Only the predictor changes within the shared planning protocol. Four layers measure open-loop prediction, cross-agent counterfactual prediction, decision quality and closed-loop mission success. Exact simulator rollouts supply the references for layers 1–3; the analysis was registered before test-set evaluation.

Across 30 trained model instances, open-loop error on U's flight plans is the strongest single correlate of mission success, while cross-agent error adds information beyond it (Fig. 5). This result depends mainly on the PETS–JEPA `flat` contrast: PETS has larger open-loop error but smaller cross-agent error and higher success. Decision quality correlates with success least. Physics with online mass estimation achieves a success rate similar to the oracle's, providing a reference for the learned models.

Our contributions are threefold:

- An open-source cooperative transport arena with hidden load mass, delayed observations and moving obstacles, a shared planner and predictor interface, and oracle and physics references.
- A four-layer evaluation linking trajectory prediction, cross-agent effects, decision quality and mission success, with exact simulator references.
- A pre-registered analysis of the relationship between prediction diagnostics and mission success across 30 trained model instances.

The code is available at [TODO: repository link; an anonymized link if the review is double-blind].

II. Related Work

Assuring learned components in aviation. Aviation regulators provide objectives and a roadmap for assuring machine-learning components [16], [17]. European guidance ties a learned surrogate's accuracy to its high-fidelity reference model [16]. We also evaluate how that accuracy translates into a planner's decisions.

Decision-aware evaluation of world models. Decision-level diagnostics assess rankings of fixed candidate actions [9], [10], [18], including in multi-agent settings [19]. Yu et al. [20] propose an evaluation ladder and a protocol with exact simulator resets and a fixed optimizer. Our protocol addresses interventional fidelity and planning utility for aircraft coupled through a shared load: it measures cross-agent counterfactual error against exact rollouts and scores the planner's execution plan rather than candidate rankings.

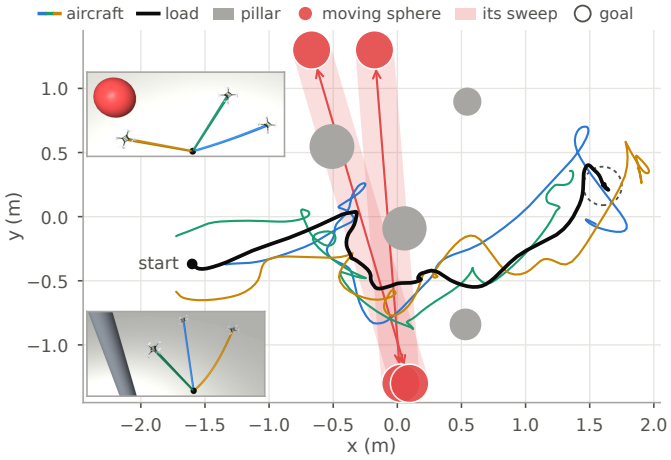


Fig. 2. Top view of one flight of the oracle U (development layout 0, moving spheres, 100 ms delay), from the start to the goal: the paths of the three aircraft, one color each, and of the load, the four pillars and the two moving spheres. Each sphere is drawn at both ends of the straight segment it shuttles along, with the strip it sweeps. Insets: the team at 3.8 s seen from above, with a moving sphere 0.5 m away (top), and at 6.4 s, with a pillar behind it (bottom); each cable is drawn in its aircraft's color.

Cooperative slung-load transport. Approaches include geometric control [12], kinodynamic planning [13], learned motor-level policies [15] and control under unknown load mass [14], [21]. Dynamics models have been compared under a common planner for a quadrotor [11], a UAV carrying a suspended payload [22], [23] and a four-UAV team in simulation [5]. We connect such controlled comparisons to cross-agent prediction and closed-loop success.

III. The Arena

The world-model comparison fixes the aircraft, onboard control, missions, observations, planner and cost. Table I summarizes the settings.

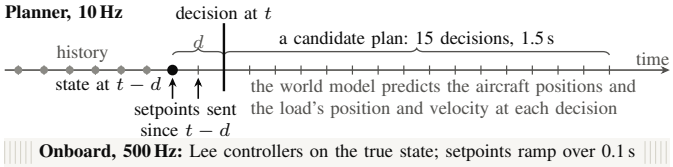
A. Task and Plant

Cables couple the aircraft through the shared load; layer 2 measures predictions of this coupling (Section IV). Hidden load mass, delayed observations and moving obstacles make the task sensitive to prediction.

Three Crazyflie 2 quadrotors from MuJoCo Menagerie [24], [25] carry a load on tension-only cables. Each uses the Crazyflie firmware's geometric tracking controller [26], based on Lee et al. [27] and compiled unchanged. The planner commands setpoints rather than thrusts. Load mass varies by layout and is hidden from every decision maker except the oracle (Section III-C).

TABLE I
TASK, PLANNING, AND EVALUATION SETTINGS

Task and environment	
Team	Three Crazyflie 2 quadrotors; one load; 0.6 m tension-only cables
Load mass	10–40 g per layout, hidden; nominal mass 25 g
Obstacles	Four pillars, with or without two moving spheres (0.25–0.45 m/s)
Mission	Goal 3.2 m ahead; time limit 20 s
Success criterion	Load within 0.15 m of the goal, below 0.2 m/s, continuously for 0.5 s
Control and planning	
Simulation/control	MuJoCo and onboard Lee controllers, both at 500 Hz
Decision rate	10 Hz
Observation delay	0, 50, 100, or 200 ms
Shared planner	MPPI; 512 candidate plans per decision
Planning horizon	15 decisions (1.5 s)
Setpoint moves	At most 0.12 m per axis
Data and evaluation	
Training budgets	500 / 2,000 episodes; 500 uses the first quarter of the same dataset
Full training set	170,304 decisions (4.7 h); 100 separate validation episodes
Main model set	5 model rows × 2 budgets × 3 new seeds = 30 models; 1.28–1.37 M parameters
Test coverage	50 layouts × 2 obstacle settings × 4 delays = 400 episodes per model
Probe states	Every fifth decision of U's test flights; 8,660 states in total
Uncertainty	95% intervals; 10,000 bootstrap resamples of the 50 test layouts



Observed: goal, pillars, setpoints sent; from $t-d$, the aircraft, load and sphere states. *Never observed:* load mass, cable tensions, controller and motor states.

Fig. 3. One decision of the shared planner, in time, as every row but the oracle U sees it.

Missions start from settled hover and succeed when the load settles at the goal beyond four pillars, with two spheres crossing the path in half the conditions (Fig. 2). Failure occurs on obstacle contact by an aircraft, load or cable; aircraft–aircraft collision; aircraft or load contact with the floor; workspace exit; or timeout.

B. Information and Latency

The planner acts at 10 Hz on states delayed by up to 200 ms; onboard Lee controllers use the true current state (Fig. 3). The predictor bridges the delay using previously sent setpoints. Observations contain states rather than images; controller and motor states remain hidden from the planner and predictor.

Two obstacle settings and four delays give eight conditions (Table I). Each layout retains its pillars and load mass across conditions and its sphere motions across delays, enabling paired comparisons. Training, validation, development and test layouts are disjoint.

C. Shared Planner and Predictor Interface

Reference predictors and learned world models use the same model predictive path integral (MPPI) planner [28], implemented in stable-worldmodel [29], with identical actions and cost. MPPI samples candidate plans, rolls them out with the predictor and commands their cost-weighted average (Fig. 1).

The predictor compensates for delay, may retain observation history, and returns aircraft positions and load position and velocity for each candidate (Fig. 3). The cost combines load distance to the goal, speed near the goal and move size with safety penalties for obstacle clearance of aircraft, load and cables, aircraft separation, load height above the floor and aircraft height above the load. All predictors share velocity-based extrapolation of sphere positions and the same MPPI random samples on each layout.

The oracle U uses the true simulator, initializing rollouts with the current undelayed state, true load mass and onboard states. Its exact predictions provide reference rollouts for all three diagnostic layers.

D. Models Under Test

Reference rows. Both physics predictors use the simulator and maintain controller and motor states in a copy of the known flight stack. From delayed observations, they replay sent setpoints to compensate for delay, as a Smith predictor does; given the true mass, they reproduce U's predictions. Nominal physics assumes a fixed mass. Online mass estimation maintains 16 instances with candidate masses from 10 to 40 g in 2 g increments, selecting the instance that best predicted the preceding 0.5 s of observations.

Learned world models. We use a joint-embedding predictive architecture (JEPA) based on LeWorldModel [30]. Conditioned on aircraft setpoints, it predicts future latent tokens from recent observations, with one token per aircraft and one for the load. A jointly trained readout decodes the states required by the cost. In `full`, all tokens attend to one another; in `no_cross`, aircraft tokens communicate only through the load token; `flat` uses one token per frame. W-ar uses the `full` backbone to predict state changes autoregressively, without latent targets. PETS (probabilistic ensembles with trajectory sampling) [31] uses an ensemble of probabilistic networks, as in four-UAV transport [5]. To match the planner's single-trajectory interface, each candidate follows the mean of one randomly selected member rather than sampled particles.

Data and training. A behavior policy collects obstacle-free formation flights between random waypoints, perturbing each aircraft's moves at MPPI's sampling scale. This covers planner-scale actions missing from the development MLP's data; the MLP retrained on these data appears in Table II, outside the primary model set. For each of five predictor variants (JEPA `full`, `no_cross`, `flat`, W-ar and PETS), up to eight configurations were compared using held-out load-position error alone. The selected configuration was trained

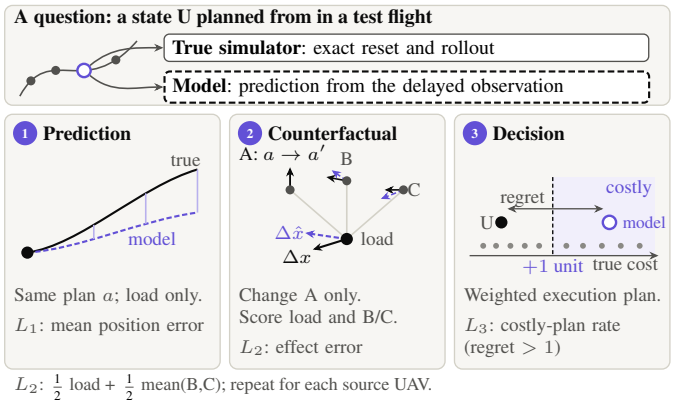


Fig. 4. How layers 1–3 score a world model. Solid: true; dashed: the model's prediction; grey dots: the candidates that U's MPPI scored. Trajectories are schematic.

with three new seeds on 2,000 episodes and the first 500 of the same dataset. Both budgets use the same configuration and number of optimization steps within each variant, yielding $5\times2\times3 = 30$ trained model instances of similar size (Table I).

Policies without the shared planner. Three policies were trained with official code and three seeds on arena layouts. CrazyMARL [15] commands rotor thrusts using each aircraft's current own state and delayed observations of the rest. It uses its own simulator with the arena's airframes, cables, obstacles and delays, nominal load mass, and the official budget of 2×10^9 steps per seed with 8,192 parallel environments (official: 16,384). TD-MPC2 [32] and DreamerV3 [33] use their official state-based settings and delayed observations for 2×10^6 online steps per seed, about twelve times the world models' data. Their reward combines load progress, a success bonus and a failure penalty. These policies lack the predictor interface and are evaluated on layer 4 only.

IV. Evaluation Layers

Layers 1–3 diagnose predictions and decisions at the same states and plans from U's test flights, rather than on held-out validation episodes. Each model receives the observations available to it inside the planner; exact simulator resets restore the true mass and onboard states to obtain reference rollouts (Fig. 4). Layer 4 measures success along each model's own closed-loop trajectories.

A. Four Layers

Let $x_{i,k}(a)$ denote the true position of the load or aircraft i after decision k of plan a, and $\hat{x}_{i,k}(a)$ its model prediction.

Layer 1: open-loop prediction. The model rolls out U's plan a. L_1 averages the load-position error $\|\hat{x}_{i,k}(a)-x_{i,k}(a)\|$ over the plan's decisions and the probe states.

Layer 2: cross-agent counterfactual prediction. We perturb one aircraft's part of a at a time to measure the predicted response of the load and teammates. Each aircraft receives eight Gaussian perturbations at MPPI's sampling scale, 0.04 m per axis and decision. For a perturbed plan a', the effect-prediction error on part i is

$$e_i = \frac{1}{15}\sum_{k=1}^{15}\left\|\Delta\hat{x}_{i,k} - \Delta x_{i,k}\right\|, \tag{1}$$

where $\Delta x_{i,k} = x_{i,k}(a') - x_{i,k}(a)$ is the true effect and $\Delta\hat{x}_{i,k}$ its prediction. Errors shared by the original and perturbed predictions cancel in this difference. L_2 averages half the load error plus half the mean error on the two unchanged aircraft, over perturbations and probe states. Median true effects on U's development states are 6.2 cm on the load and 2.7 cm on another aircraft.

Layer 3: decision quality. Each model predicts the same candidates scored by U's MPPI. The shared cost and frozen MPPI update produce a cost-weighted execution plan. Regret is its true cost minus that of U's execution plan; a decision is costly if regret exceeds one cost unit, approximately a 1.7 cm loss of safety margin or 0.6 m less progress. This measures an error's decision cost rather than its magnitude [34]. L_3 is the costly-decision rate on probe states where U's own weighted plan is within one cost unit of its best candidate: 89% of states, identical across models.

Layer 4: mission success. Each world model runs the test episodes in the shared planner, and each policy runs them independently. J is the success rate. This layer captures errors accumulating along each model's own flight.

B. Pre-registered Analysis

The version-tagged analysis plan, fixed before test-set evaluation, tests whether L_2 adds information about J beyond L_1 across the 30 model instances. The primary statistic is the partial Spearman correlation, computed from pairwise rank correlations ρ:

$$\rho(L_2, J \mid L_1) = \frac{\rho(L_2,J) - \rho(L_2,L_1)\,\rho(L_1,J)}{\sqrt{\big(1-\rho(L_2,L_1)^2\big)\big(1-\rho(L_1,J)^2\big)}}. \tag{2}$$

The 95% interval resamples test layouts with their episodes and probe states, recomputing all models on each resample to preserve pairing across models and layers. The 30 model instances remain fixed (Table I).

An interval entirely below zero supports the claim; one containing zero leaves it unestablished. All 30 models completed the protocol and enter the analysis. Simulated power is 67–79% for a partial correlation of 0.43–0.48.

Secondary analyses report intervals without a decision rule: layer 3 conditioned on layer 1, each layer's plain correlation with J, the primary statistic by condition, data-budget comparisons and JEPA-variant comparisons paired by seed and budget. The released analysis plan and code specify cost weights, stochastic-predictor scoring and development-set choices.

V. Results

A. Benchmark Results

The best learned world models approach nominal physics in mission success. Oracle U's failures are nearly all

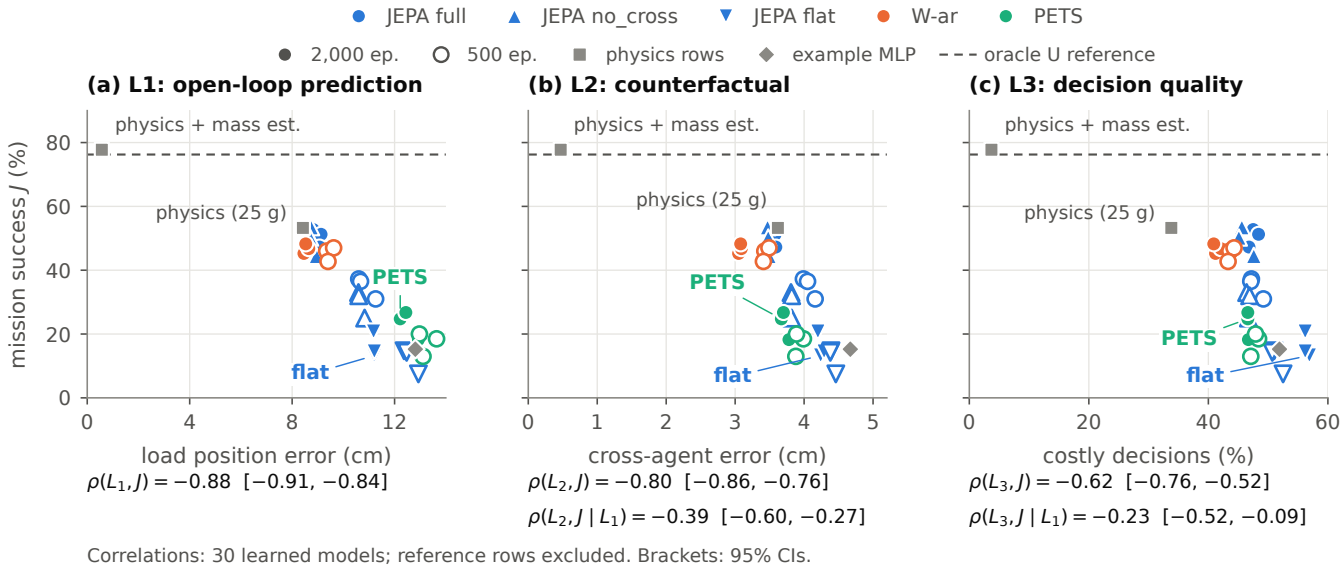


Fig. 5. Mission success against layers 1–3, one point per model instance. PETS has larger open-loop error than JEPA `flat`, but smaller cross-agent error and higher success. Correlations use the 30 learned instances, excluding reference rows and the example MLP. Panels report Spearman correlations and 95% intervals; the middle and right also report partial correlations given L_1, (2). Marker encodings are defined in the legend.

timeouts despite exact predictions (Table II; Fig. 6). Physics with online mass estimation achieves a similar success rate. Nominal-mass physics succeeds in about half the missions, with aircraft–aircraft collisions its most frequent failure, concentrated on heavy loads. At 2,000 training episodes, JEPA `full`, `no_cross` and W-ar achieve success rates similar to nominal physics; PETS and `flat` succeed in at most a quarter of missions. Learned-model failures are mostly collisions. Among independent policies, CrazyMARL succeeds in about two missions in five; TD-MPC2 and DreamerV3 have no successes.

B. Findings

Cross-agent error adds information about mission success beyond open-loop error. Open-loop error on U's plans is the strongest single correlate of success (Fig. 5). Controlling for it, cross-agent error remains negatively associated with success across the 30 model instances (partial Spearman correlation -0.39, 95% interval $[-0.60, -0.27]$), supporting the pre-registered claim. PETS illustrates the additional information: it has larger open-loop error than JEPA `flat`, but smaller cross-agent error and higher success (23% versus 16% at 2,000 episodes). In an exploratory check, removing either row makes the interval include zero (Fig. 7), indicating that this contrast is central to the pooled result.

The clearest layer 2 associations occur with static obstacles and short delays. Partial correlations are negative in all eight conditions, with intervals below zero only for pillars alone at delays up to 100 ms (Fig. 7). Layer 1 remains closely associated with success across conditions.

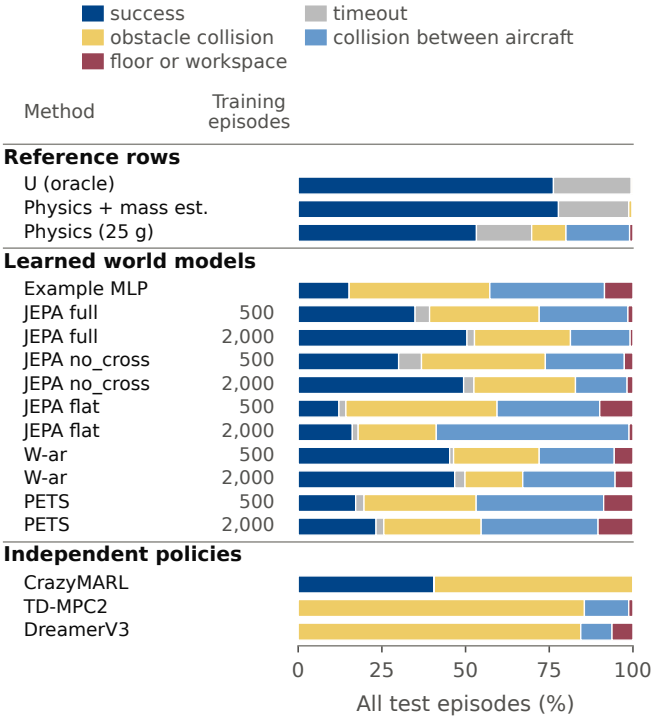
Decision quality has the weakest correlation with mission success. Learned models make costly decisions at roughly half the probe states. At 2,000 episodes, PETS and JEPA `full` have similar costly-decision rates, yet `full` succeeds more than twice as often (Fig. 5). Layer 3's partial correlation with success given layer 1 is also negative, with the association again dependent on PETS and `flat`.

Entity tokens improve JEPA performance; direct aircraft attention adds little. JEPA `full` and `no_cross` achieve two and a half to three times `flat`'s success and lower

TABLE II
BENCHMARK RESULTS ACROSS EIGHT TEST CONDITIONS

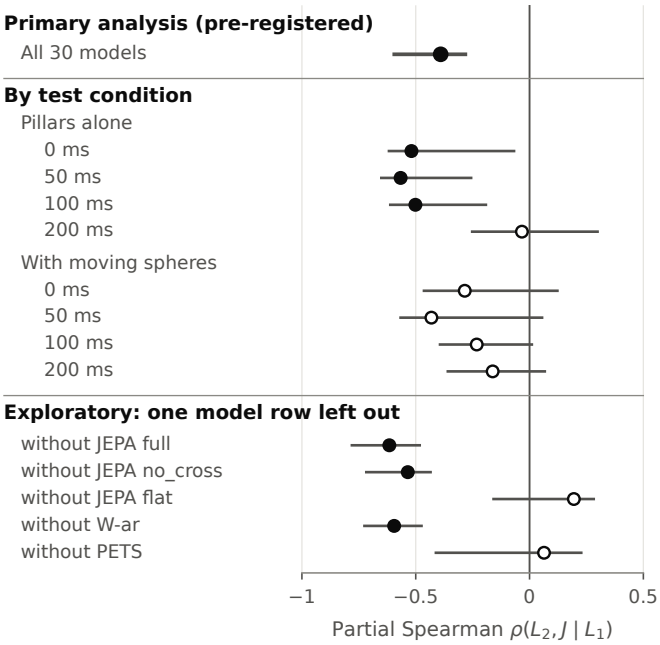
Method	Train ep.	J (%) ↑ [95% CI]	L_1 (cm) ↓	L_2 (cm) ↓	L_3 (%) ↓
Reference rows, shared MPPI					
U (oracle)	–	76.3 [65.0, 86.8]	0	0	0
Physics + mass est.	–	77.8 [67.5, 87.0]	0.57	0.47	3.7
Physics (25 g)	–	53.3 [42.5, 64.3]	8.42	3.62	33.8
Learned world models, shared MPPI					
Example MLP	–	15.3 [9.3, 22.3]	12.81	4.67	51.9
JEPA `full`	500	34.9 [26.5, 43.6]	10.84	4.07	47.8
JEPA `full`	2,000	50.4 [40.3, 60.3]	9.01	3.56	47.6
JEPA `no_cross`	500	30.0 [22.6, 37.8]	10.67	3.82	46.7
JEPA `no_cross`	2,000	49.4 [39.8, 59.0]	8.80	3.48	46.1
JEPA `flat`	500	12.2 [7.1, 18.3]	12.59	4.41	51.4
JEPA `flat`	2,000	16.2 [11.4, 21.5]	11.20	4.25	56.4
W-ar	500	45.3 [36.2, 54.3]	9.46	3.44	43.7
W-ar	2,000	46.8 [37.8, 55.5]	8.55	3.07	41.4
PETS	500	17.2 [11.4, 23.7]	13.24	3.92	47.8
PETS	2,000	23.3 [16.4, 30.9]	12.53	3.72	46.6
Policies without the shared planner					
CrazyMARL	–	40.6 [30.0, 51.5]	–	–	–
TD-MPC2	–	0.0[a]	–	–	–
DreamerV3	–	0.0[a]	–	–	–

400 test episodes per model. For three-seed methods, J pools 1,200 episodes; L_1–L_3 are seed means. Brackets: 95% bootstrap CIs over 50 layouts. Train ep.: offline training episodes; –: not reported or not applicable. [a]Zero successes across all three seeds.



400 flights per model; 1,200 for three pooled seeds.

Fig. 6. Test outcomes by method and training budget. Oracle U's failures are nearly all timeouts; learned models and policies fail mainly by collisions. Floor or workspace denotes aircraft or load contact with the floor, or workspace exit. Three-seed rows pool 1,200 episodes; other rows contain 400.



Points: estimates; lines: 95% bootstrap intervals.
Filled: interval below 0; open: interval includes 0.

Fig. 7. The partial Spearman correlation of layer 2 with success given layer 1, (2), with 95% bootstrap intervals over layouts. Top: the pre-registered test across all 30 learned models. Middle: the eight test conditions. Bottom: an exploratory check outside the analysis plan, leaving out one model row's six models at a time. Filled markers indicate intervals entirely below zero; open markers indicate intervals containing zero.

cross-agent error (Table II). At 2,000 episodes, aircraft–aircraft collisions end more than half of `flat`'s missions but fewer than one in five of `full`'s (Fig. 6). Despite slightly lower cross-agent error, `no_cross` has slightly but measurably lower success than `full`.

More data improves prediction more consistently than mission success. Increasing the budget from 500 to 2,000 episodes lowers L_1 and L_2 for every predictor variant. Success increases for `full`, `no_cross` and PETS; the two entity-token JEPA variants rise from about a third to about half of missions. Changes in `flat` and W-ar success are not detectable.

VI. Discussion

What each layer tells. Trajectory and cross-agent errors provide complementary diagnostics, with the latter's additional association here driven mainly by PETS and `flat`. Neither replaces closed-loop evaluation. Layer 3 measures single decisions at U's states; replanning 0.1 s later may correct a costly decision, while layer 4 captures errors accumulated on a model's own trajectory. In this arena, aircraft interact only through cables and the load, apart from contact; aerodynamic interaction is absent. That load-mediated path remains available to `no_cross`.

[TODO: The shared planner sets the ceiling.] [TODO: U fails 95 of 400 test episodes with exact predictions, so part of every row's failures belongs to the search, not the model; layers 2 and 3 separate the two. A stronger planner is future work, and so is an inference-compute axis (varying the MPPI sample budget), which the frozen analysis does not include.]

[TODO: Online model-based RL did not solve the task.] [TODO: TD-MPC2 and DreamerV3 succeeded in none of 2,400 test episodes; reported as is, outside the primary analysis.]

[TODO: From state to perception, and from simulation to flight.] [TODO: The arena observes states; onboard perception (cameras, for example) and real cables are open steps. One task, one planner and 30 models: say what generalizes and what does not. Layers 1–3 ask at U's states, and only layer 4 sees a model's own (Section IV-A); say more here only if space allows.]

VII. Conclusion

[TODO: Three or four sentences: the question, the arena and its layers, the answer, and what it means for validating learned models before flight. Code and data availability.]

Acknowledgment

[TODO: Funding and acknowledgments, or delete this section.]

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